

AFTER — both changes together

When: M26S1J21Q55 Topic: Vector algebra toolkit (cf. A14)

Given: $\vec{a} = -\hat{i} + 2\hat{j} + 2\hat{k}$, $\vec{b} = 8\hat{i} + 7\hat{j} - 3\hat{k}$, $\vec{c}$ with $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = 4$.

Find: $|\vec{a} + \vec{c}|^2$.

UF–S Solves.

1. $\vec{a} \times \vec{c} = \vec{b}$ in components gives three scalar equations in c_1, c_2, c_3 .
2. Only two are independent (the third follows automatically, since $\vec{a} \cdot \vec{b} = 0$).
3. Set $c_3 = t$: the two independent equations give c_1, c_2 in terms of t .

Predict: substitute into $\vec{c} \cdot (1, 1, 1) = 4$ to solve for t , then compute $\vec{c}$.

What this is asking: $|\vec{a} + \vec{c}|^2$ is the squared length of the vector sum $\vec{a} + \vec{c}$ — a single number measuring how far the point $\vec{a} + \vec{c}$ sits from the origin, once $\vec{c}$ itself has been pinned down by the two given conditions.

Foundation — Underdetermined cross-product. $\vec{a} \times \vec{c} = \vec{b}$ (with $\vec{a} \cdot \vec{b} = 0$) determines $\vec{c}$ only up to adding a multiple of $\vec{a}$.

- Compute $\vec{a} \times \vec{c} = (2c_3 - 2c_2, 2c_1 + c_3, -c_2 - 2c_1)$ and match to $\vec{b} = (8, 7, -3)$: gives $c_3 - c_2 = 4$ and $2c_1 + c_3 = 7$.
- Set $c_3 = t$: then $c_2 = t - 4$ and $c_1 = (7 - t)/2$.
- Impose $\vec{c} \cdot (1, 1, 1) = 4$, i.e. $c_1 + c_2 + c_3 = 4$: solves to $t = 3$, so $\vec{c} = (2, -1, 3)$.
- $\vec{a} + \vec{c} = (1, 1, 5) \Rightarrow |\vec{a} + \vec{c}|^2 = 27$.

When: M26S1J21Q64 Topic: A point and a line (cf. A6+A14)

Given: (α, β, γ) is the foot of the perpendicular from $(5, 4, 2)$ to the line $\vec{r} = (-\hat{i} + 3\hat{j} + \hat{k}) + \lambda(2\hat{i} + 3\hat{j} - \hat{k})$. **Find:** the projection of $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ onto $6\hat{i} + 2\hat{j} + 3\hat{k}$.

What this is asking: This asks for the length of the shadow that one specific point — the closest point on a given line to $(5, 4, 2)$ — casts onto a chosen direction $(6, 2, 3)$.

UF–S Solves.

1. Every point on the line is $\vec{r}(\lambda) = (-1 + 2\lambda, 3 + 3\lambda, 1 - \lambda)$.
2. The foot of the perpendicular from $P = (5, 4, 2)$ is the λ where $\vec{r}(\lambda) - P$ is perpendicular to the line's direction $(2, 3, -1)$.
3. Perpendicular means their dot product is zero.

Predict: one linear equation in λ ; solve it, get the foot, then project onto $(6, 2, 3)$.

Foundation — Foot-of-perpendicular. On a line $\vec{r}(\lambda) = A + \lambda\vec{d}$, the foot from P is the λ solving $(\vec{r}(\lambda) - P) \cdot \vec{d} = 0$.

- Compute $\vec{r}(\lambda) - P = (2\lambda - 6, 3\lambda - 1, -\lambda - 1)$.
- Dot with the direction $(2, 3, -1)$ and set to zero: $14\lambda - 14 = 0 \Rightarrow \lambda = 1$.
- Foot = $(-1 + 2, 3 + 3, 1 - 1) = (1, 6, 0)$.
- Project onto $(6, 2, 3)$: $\frac{1(6) + 6(2) + 0(3)}{\sqrt{36 + 4 + 9}} = \frac{18}{7} \Rightarrow 18/7$.

When: M26S2J21Q3 Topic: Two lines together (cf. A14)

Given: L_1 through $(2, 6, 7)$ parallel to $(-3, 2, 4)$; L_2 through $(4, 3, 5)$ parallel to $(2, 1, 3)$; L_3 parallel to $(-3, 5, 16)$ meets L_1 at C , L_2 at D . **Find:** $|\vec{CD}|^2$.

UF–S Solves.

1. Write C and D as points on L_1, L_2 using their own parameters: $C = (2, 6, 7) + t(-3, 2, 4)$, $D = (4, 3, 5) + s(2, 1, 3)$.
2. $\vec{CD} = D - C$ must be parallel to L_3 's direction $(-3, 5, 16)$ — three scalar equations (one per coordinate) in the two unknowns t, s , over-determined but consistent since L_3 genuinely exists.

Predict: solve for t, s from the parallel condition, then compute C, D and the distance directly.

What this is asking: This is asking for the squared straight-line distance between two points: C , where a line parallel to a given direction meets L_1 , and D , where that same line meets L_2 .

Foundation — Parametrised-points-on-two-lines. Two points constrained to lie on two given lines, with their connecting vector forced to a third fixed direction, is a linear system in the two line-parameters — no geometric guessing needed.

- $D - C = (2 + 3t + 2s, -3 - 2t + s, -2 - 4t + 3s)$ must be proportional to $(-3, 5, 16)$.
- Solving the resulting equations gives $t = -3, s = 2$ (checked consistent across all three coordinates).
- $C = (11, 0, -5)$, $D = (8, 5, 11)$.
- $|\vec{CD}|^2 = (-3)^2 + 5^2 + 16^2 = 9 + 25 + 256 \Rightarrow \boxed{290}$.